

Airbnb Superhost Classification Analysis

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Machine Learning Approach to Identifying High-Performance Hosts in New York City



ROC-AUC Score
0.99



Dataset Size
~36,000



F1 Score
0.71

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Project Overview



Industry Context

This project focuses on the **short-term rental industry**, specifically Airbnb listings in New York City. On Airbnb, hosts can obtain **Superhost status** awarded to hosts who consistently provide excellent service and maintain high guest satisfaction.



Project Objective

Build a **machine learning model** that can classify whether a host is likely to become a Superhost based on listing characteristics, host behavior, and location-related attributes. Understanding these factors helps identify patterns associated with high-quality hosting performance.



Business Value

Identifying these patterns allows platforms like Airbnb to **encourage better host practices**, improve overall service quality, and strengthen the trust ecosystem between guests and hosts. The model provides insights into key factors influencing Superhost status.



Superhost Benefits



Higher Visibility

Superhost listings receive priority placement in search results and are prominently featured on the platform.



Increased Bookings

The Superhost badge builds immediate trust with potential guests, leading to significantly higher conversion rates.



Guest Trust

The badge signals reliability and quality, reducing booking friction and increasing guest confidence.

Key Success Metrics

4.8+

Avg Rating

90%+

Response Rate

10+

Stays/Year

1%

Cancel Rate

Dataset Overview & Key Metrics



Total Listings

35,953



Superhosts

6,831



Non-Superhosts

29,122



Superhost Rate

19.0%

Dataset Features (18 Variables)

Behavioral (5)

- host_response_rate
- availability_365
- number_of_reviews
- review_scores_rating
- instant_bookable

Location (2)

- latitude, longitude
- neighbourhood_group

Host Attributes (3)

- host_identity_verified
- host_verifications_count
- host_tenure_days

Structural (8)

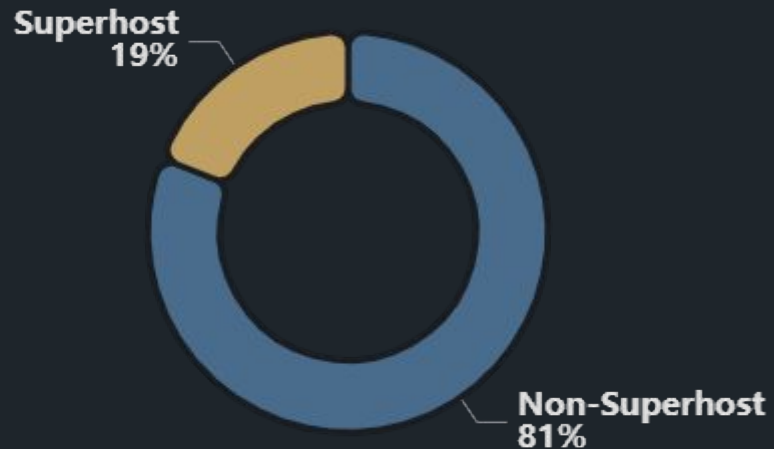
- accommodates, bathrooms
- bedrooms, beds
- min/max nights
- amenities_count

Top 5 Key Features

- 1** **host_response_rate**
▲ Increases Superhost probability
- 2** **review_scores_rating**
▲ Increases Superhost probability
- 3** **amenities_count**
▲ Increases Superhost probability
- 4** **number_of_reviews**
▲ Increases Superhost probability
- 5** **maximum_nights**
▼ Decreases Superhost probability

Superhost Distribution Analysis

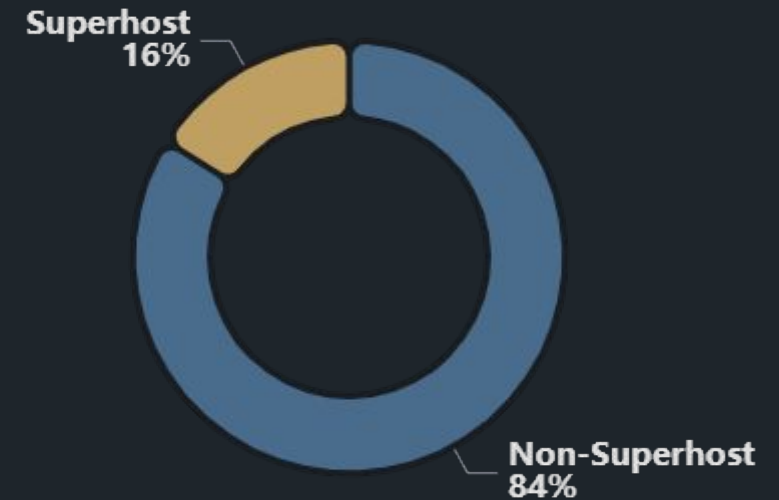
 Overall Proportion by Total Listings



 Insight

The overall proportion indicates a **class imbalance** between Superhosts and non-Superhosts, with only about **19%** of hosts classified as Superhosts. However, this figure is based on **unaggregated listing-level data**, meaning the same host may be counted multiple times if they have multiple listings.

 Proportion by Host ID (Aggregated)

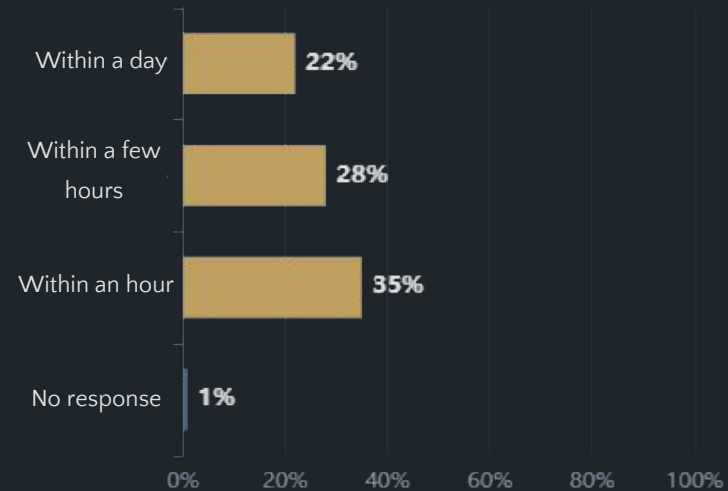


 Insight

The proportion aggregated by **host_id** drops to **16%**, indicating that hosts with very large numbers of listings (100–1000) are typically **not Superhosts**. At the listing level, these hosts can skew the overall proportion due to their large listing volume.

Behavioral Factors Analysis

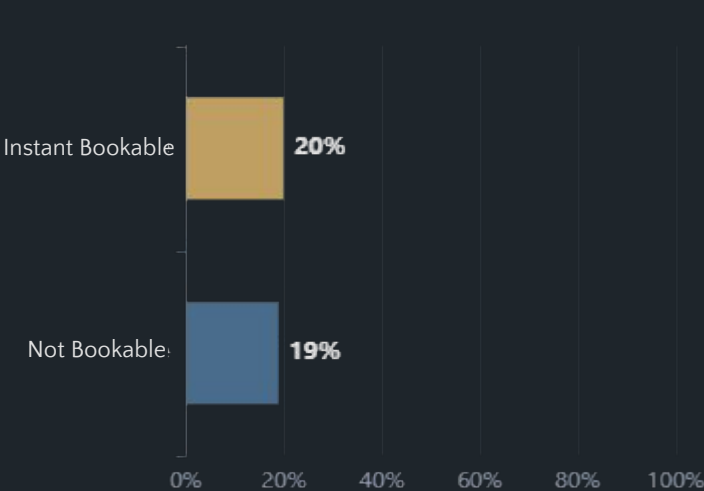
🕒 Host Response Time



💡 Insight

Hosts with shorter response times achieve **35%** Superhost rate. The large "no response" category (**1%** rate) significantly lowers overall proportion.

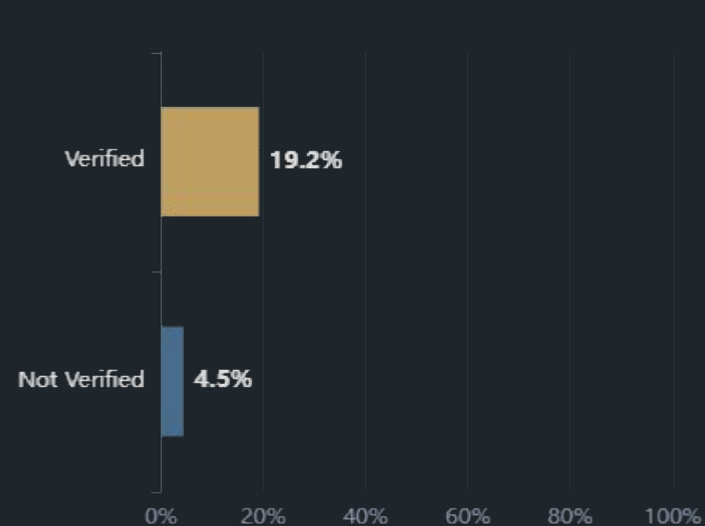
⚡ Instant Bookable



💡 Insight

Minimal difference in Superhost rates. Instant booking adoption is **4x lower** than manual approval, indicating host preference for control.

🛂 Host Identity Verified



💡 Insight

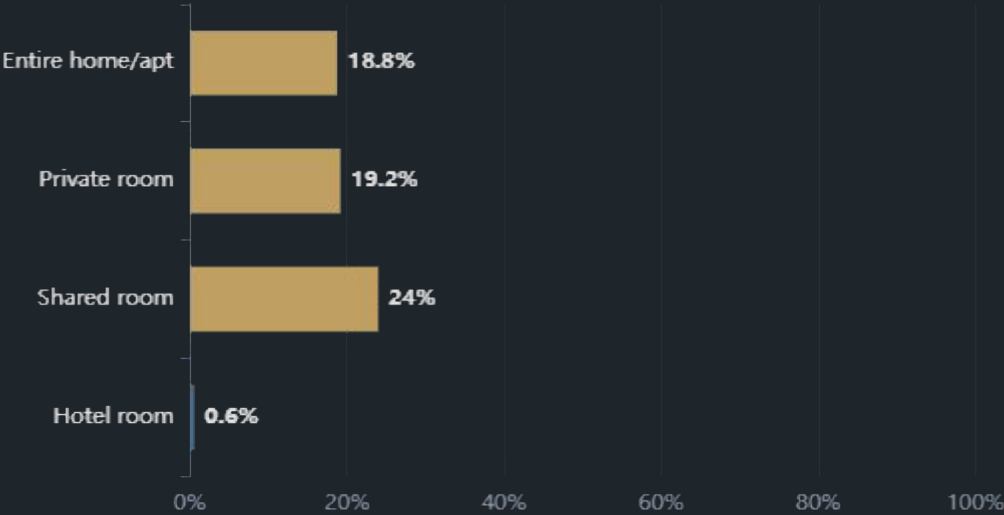
Non-verified hosts show **4.5%** vs. **19.2%** for verified. However, non-verified hosts represent <1,000 listings with minimal overall impact.

📋 Key Behavioral Insights

Behavioral factors reveal that **host responsiveness** is the strongest differentiator, with response time showing dramatic impact on Superhost likelihood. Instant booking and identity verification show weaker predictive power, suggesting that **active engagement** matters more than structural account features.

Structural & Location Factors

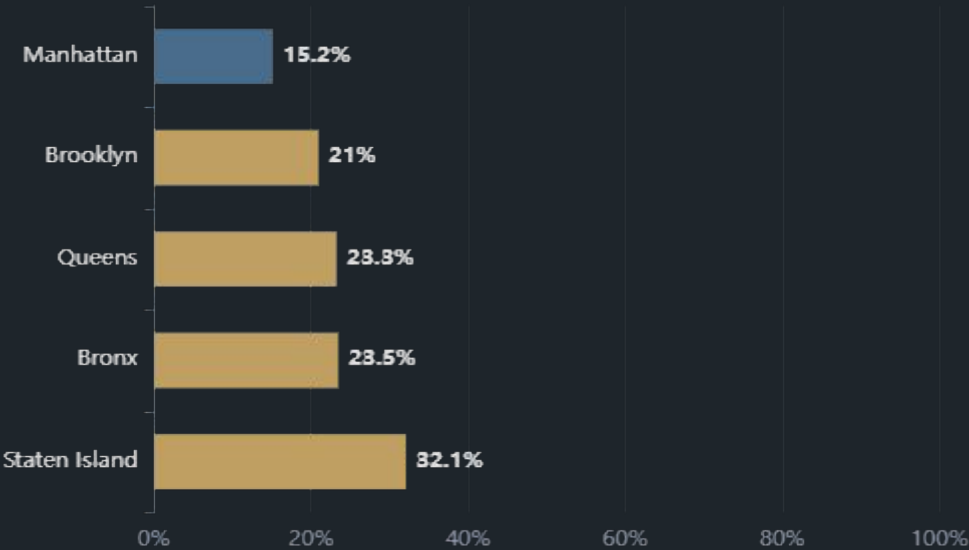
🏠 Room Type Distribution



💡 Insight

Entire home/apt and Private room dominate with 15,000+ listings each, both showing -19% Superhost rate. Hotel room has 0.6% rate but small volume.

📍 Borough Distribution



💡 Insight

Manhattan has highest listings but lowest rate (15.2%). Brooklyn (21%) and Queens (23.3%) perform better.

Entire Home/Apt

-15,500
18.8%

Private Room

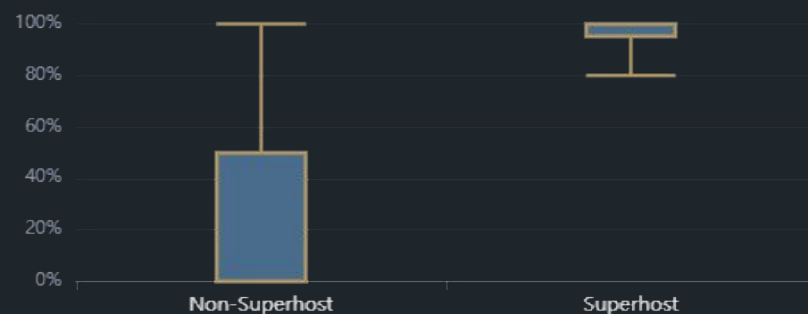
-15,200
19.2%

Manhattan Listings

-14,000
15.2%

Numerical Features: Key Differentiators

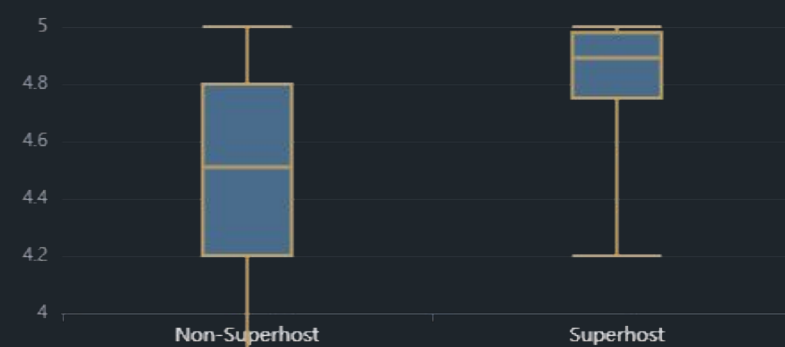
🔍 Host Response Rate



💡 Insight

Dramatic difference: Non-Superhost median 0% vs. Superhost median 100%. Shows communication reliability is critical.

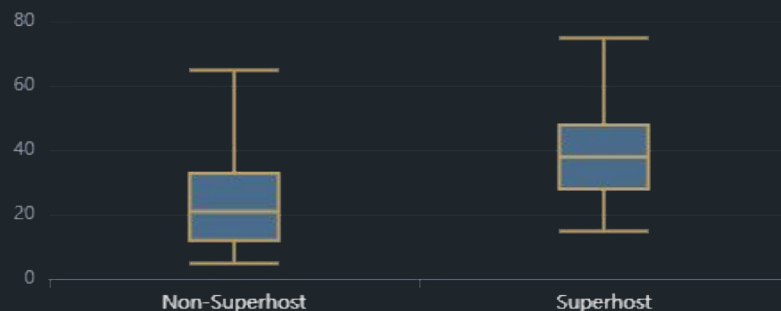
★ Review Scores Rating



💡 Insight

Non-Superhost median 4.51 vs. Superhost 4.89. Superhosts consistently near maximum 5.0.

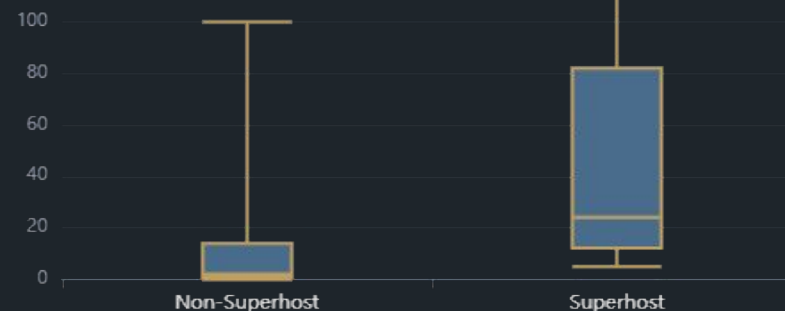
🏠 Amenities Count



💡 Insight

Non-Superhost median 21 vs. Superhost 38 amenities. Upper quartile: 48 vs. 33.

💬 Number of Reviews

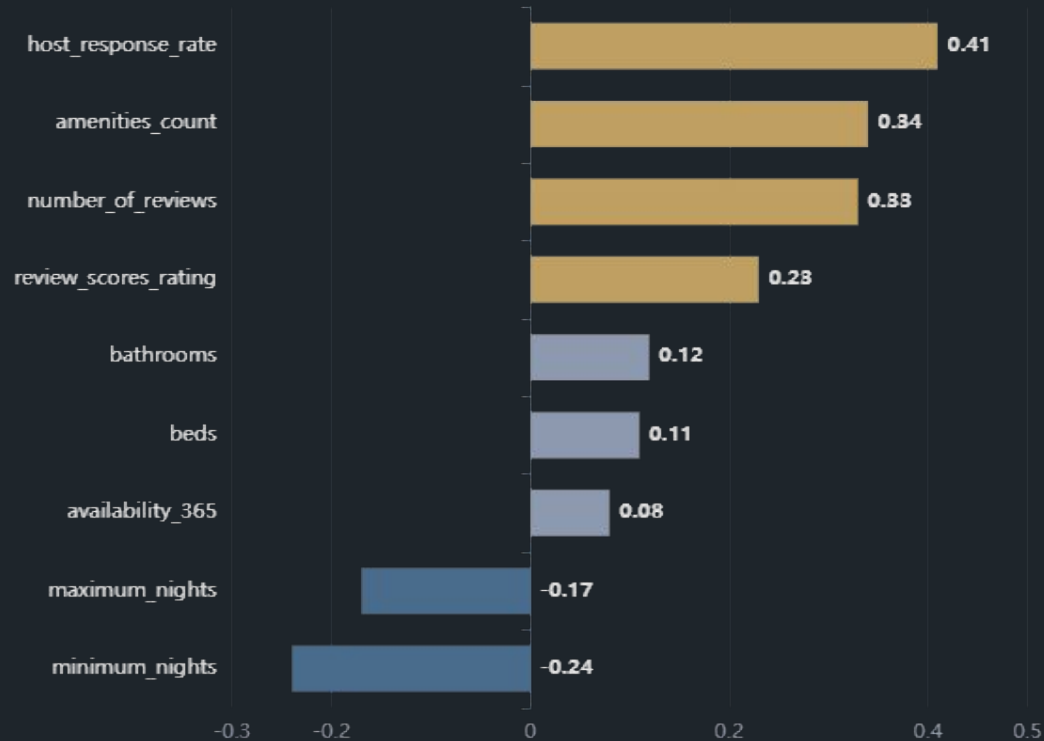


💡 Insight

Non-Superhost median 2 vs. Superhost 24 reviews. Upper quartile: 82 vs. 14.

Correlation Analysis Insights

Feature-to-Target Correlation



↑ Strongest Positive Correlations

host_response_rate	0.41
amenities_count	0.34
number_of_reviews	0.33
review_scores_rating	0.23

↓ Negative Correlations

minimum_nights	-0.24
maximum_nights	-0.17

💡 Key Insight

No single feature shows a very strong correlation with the target, indicating that **Superhost status is influenced by a combination of multiple factors** rather than a single dominant variable. This suggests a **multivariate approach** is necessary for accurate prediction.

Model Performance Comparison

🏆 Performance Metrics (4:1 Train-Test Split)

Dummy Baseline	Logistic Regression	Random Forest	LR (Tuned)	RF (Tuned) ★
0.820	0.892	0.942	0.892	0.949
Accuracy	ROC-AUC	ROC-AUC	ROC-AUC	ROC-AUC

💡 Key Findings

The baseline model achieves 0.82 accuracy by always predicting the majority class but has 0% recall. Logistic Regression improves recall to 0.87 but precision only 0.46. Random Forest provides better balance with precision 0.57, recall 0.89, F1 0.70. Tuned Random Forest achieves best performance: F1 0.71, ROC-AUC 0.949.

📊 Confusion Matrix

Random Forest

	Non Superhost	Superhost
Non Superhost	TN 5109	FP 851
Superhost	FN 142	TP 1151
	Non Superhost	Superhost

Prediction Label

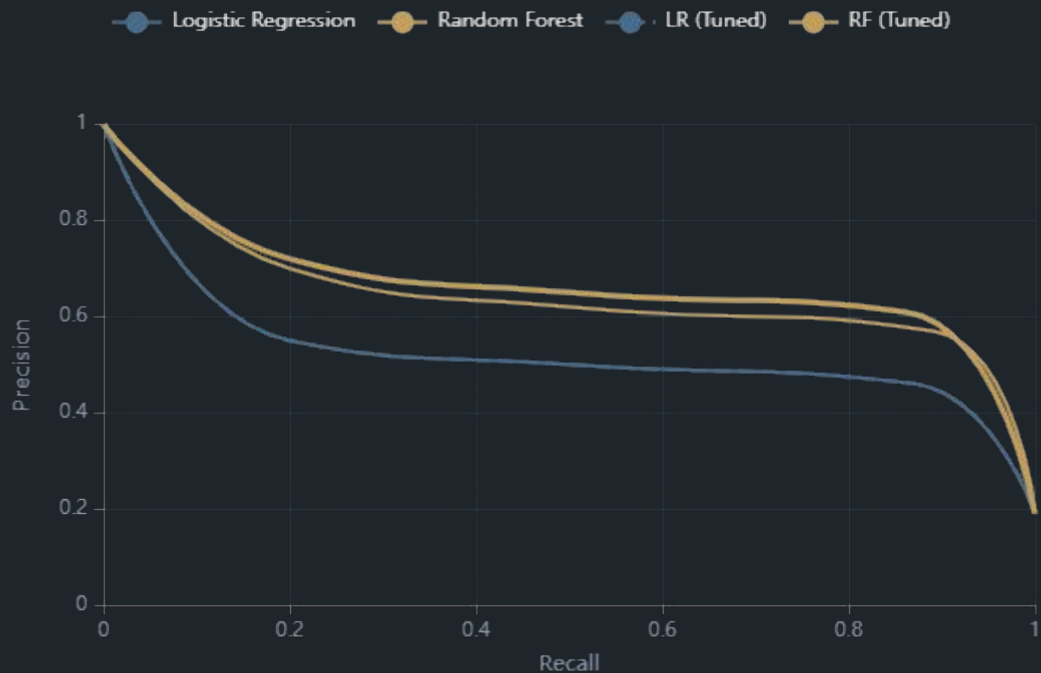
RF (Tuned)

	Non Superhost	Superhost
Non Superhost	TN 5240	FP 720
Superhost	FN 175	TP 1118
	Non Superhost	Superhost

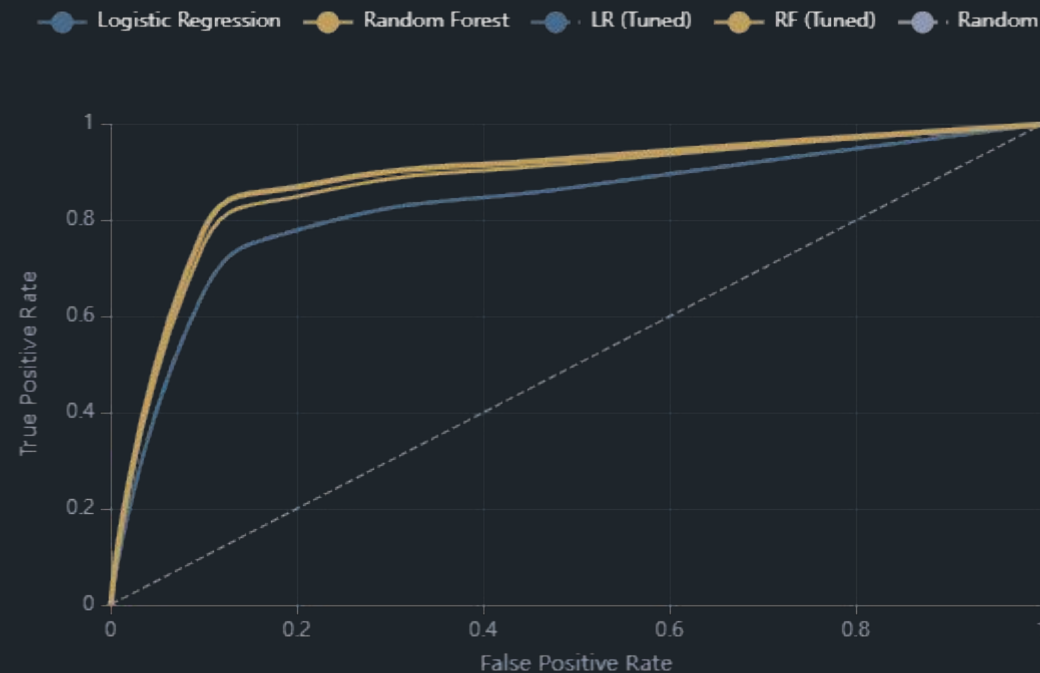
Prediction Label

Model Evaluation

Precision-Recall Curve



ROC Curve



Baseline Model

Contains only **TN and FN** due to most-frequent strategy always predicting Non-Superhost. All actual Superhosts appear as false negatives.

Logistic Regression

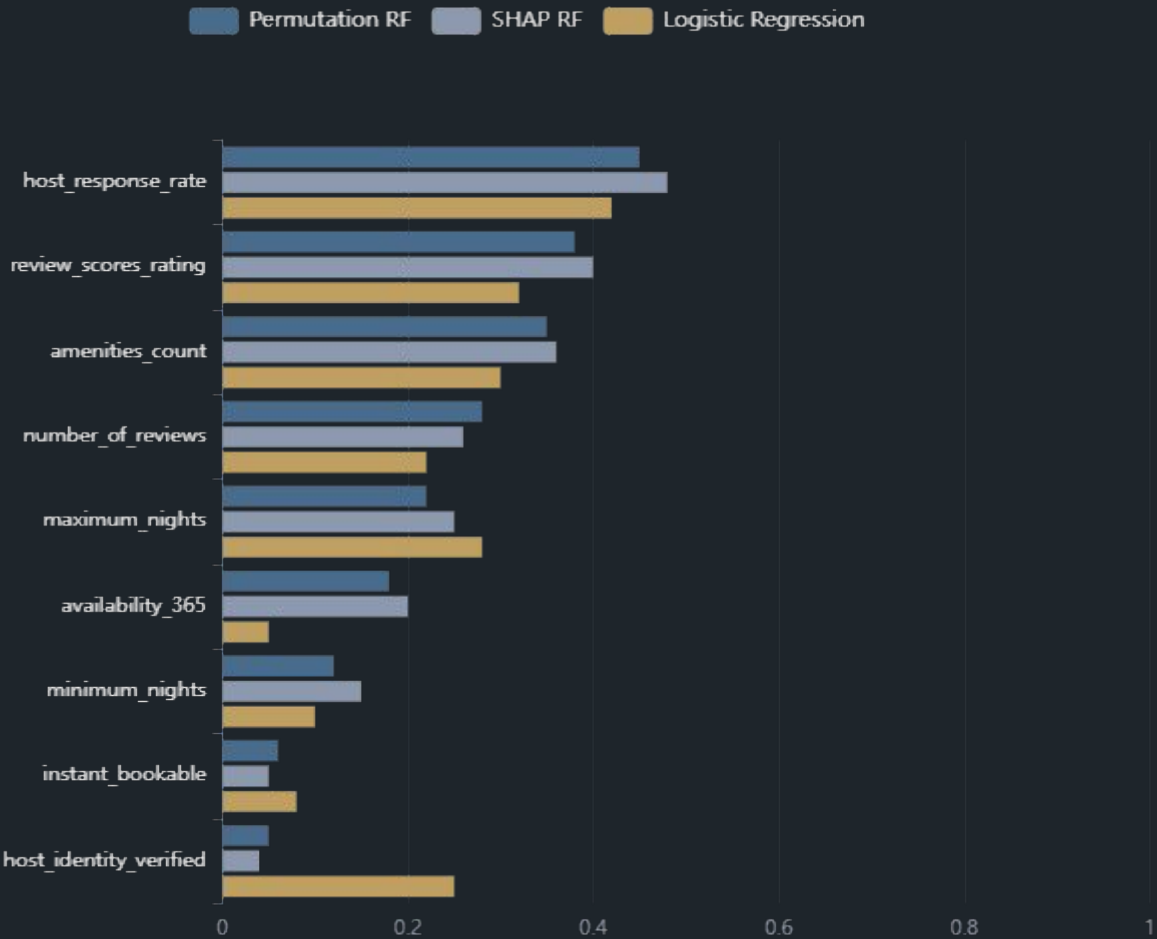
Performance remains **nearly identical** after tuning (1,130 TP, 163 FN). Default parameters were near-optimal for this dataset.

Random Forest

Shows **trade-off** after tuning: TP decreases (1,151→1,118) but TN increases (5,109→5,240). Model becomes more conservative.

Feature Importance Analysis

Multi-Method Comparison (Normalized)



Top 5 Consistent Features

- 1** **host_response_rate**
Behavioral
- 2** **review_scores_rating**
Behavioral
- 3** **amenities_count**
Structural
- 4** **number_of_reviews**
Behavioral
- 5** **maximum_nights**
Structural

Method Differences

Logistic Regression emphasizes **OHE features** (hotel_room, shared_room). Random Forest distributes importance across **correlated features**, making binary variables less prominent.

Key Findings & Strategic Implications

1

Host Behavior Dominates

Host behavior metrics (response rate, review scores) are the strongest predictors of Superhost status, emphasizing **service quality over structural factors**. Platforms should prioritize encouraging host responsiveness and guest satisfaction initiatives.

2

Class Imbalance Insights

19% Superhost rate at listing level drops to **16%** when aggregated by host_id. High-volume hosts (100-1000 listings) are typically **not Superhosts**, suggesting quality vs. quantity trade-off in hosting strategies.

3

Model Selection

Random Forest outperforms Logistic Regression (ROC-AUC 0.949 vs. 0.892) with better precision-recall balance. The tuned RF model achieves F1 0.71, making it the **recommended model** for production deployment.

4

Core Predictive Features

Five features consistently drive predictions: **host_response_rate**, **review_scores_rating**, **amenities_count**, **number_of_reviews**, **maximum_nights**. These capture host behavior, listing quality, and structural characteristics.

Strategic Implications for Airbnb

Platform Focus

Encourage host responsiveness through notifications and incentives

Host Development

Educate hosts on amenity offerings and guest experience optimization

Policy Insights

Longer maximum nights negatively impact Superhost likelihood

Conclusion

This analysis successfully identified **key factors that distinguish Superhosts** from regular hosts in NYC's Airbnb market.

The tuned Random Forest model achieved strong predictive performance (**ROC-AUC: 0.949**, **F1: 0.71**), demonstrating that machine learning can effectively support platform decision-making. The findings emphasize that **host behavior and service quality metrics** outweigh structural and location factors in determining Superhost status, providing actionable insights for platform optimization and host development programs.



35,953

Listings Analyzed



5

ML Models Evaluated



0.949

Best ROC-AUC Score



March 2026



Machine Learning Classification Project